

Predicting E-Commerce Customer Churn

A From-Scratch Data Preprocessing and Statistical Feature Selection Pipeline

Course: Machine Learning Practical

Problem Type: Binary Classification (Target: Churn $\in \{0, 1\}$)

Source: Kaggle E-Commerce Churn Dataset (5,630 Records)

Repository: github.com/DrishtiNigam1928/CUSTOMER_CHURN

1. PROJECT TEAM & TASK OWNERSHIP MATRIX

Student Name	Roll Number	Assigned Tasks & Module Contributions
Anmol Pandey	06	B1 (Data Exploration), E1 (Label Encoding), E2 (One-Hot Encoding), F2 (Z-Score Outliers), H2 (Standardization Engine), K (Data Leakage Architecture), L (Pipeline Integration), M4 (ANOVA F-Test), M6 (Technique Matrix), M7 (Feature Selection Table), M9 (Leakage Audit), N (Audit Summary)
Drishti Nigam	56	A2 (Dataset Profiling), B2 (Descriptive Statistics Engine), D1 (ID-Scan Duplicate Detection), D2 (Inconsistent Text Cleaning), F1 (IQR Outlier Detection), H1 (Min-Max Normalization), L (Pipeline Design), M1 (Variance Threshold), M3 (Chi-Square Test of Independence), M9 (Supervised Leakage Verification), N (Comparison Table)
Abhishek Saini	02	C1 (Missing Value Analysis), C2 (Skew-Based Imputation), F3 (Outlier Treatment Logic), G (Log Transformation), I (Distribution Histograms), J (From-Scratch 80/20 Train-Test Split), L (Pipeline Sequencing), M2 (Pearson Correlation), M8 (Feature Transition Audit), M9 (Feature Selection Leakage Verification), N (Final Synthesis)

2. PROBLEM DEFINITION & OPERATIONAL OBJECTIVES (TASKS A1 & A2)

In the e-commerce sector, customer churn (attrition) directly undermines recurring revenue. Acquiring new customers costs 5× to 7× more than retaining existing users. Predicting churn allows proactive customer-success outreach, tailored loyalty incentives, and logistical improvements.

Core Philosophy & Workflow Architecture

To avoid black-box machine learning, this project executes an end-to-end mathematical pipeline developed entirely from first principles:

Theory → **Mathematical Formula** → **From-Scratch Python Implementation** → **Empirical Output** → **Statistical Interpretation**

All preprocessing parameters are strictly fitted on the training split (4,504 records) and applied downstream to prevent data leakage.

3. DATASET DESCRIPTION & VARIABLE ARCHITECTURE

The dataset contains **5,630 raw observations** (partitioned into **4,504 training rows** and 1,126 testing rows via an 80/20 random index split). The initial schema consists of **20 columns** (1 Primary Key, 1 Target Variable, and 18 Input Features: 13 Numerical, 5 Categorical).

Feature Name	Data Type	Variable Role	Description & Operational Meaning
CustomerID	Integer	Identifier	Unique account identifier for primary key tracking.
Churn	Binary Integer	Target (Output)	0 = Customer Retained, 1 = Customer Churned (Attrition event).
Tenure	Numerical (Float)	Input	Duration (in months) of active customer tenure on the platform.
PreferredLoginDevice	Categorical	Input	Primary login device (Phone, Mobile Phone, Computer).
CityTier	Numerical (Int)	Input	Urban classification tier of customer residence (1, 2, or 3).

Feature Name	Data Type	Variable Role	Description & Operational Meaning
WarehouseToHome	Numerical (Float)	Input	Distance (km) from closest fulfillment center to delivery location.
PreferredPaymentMode	Categorical	Input	Primary payment method (Debit Card, Credit Card, E Wallet, UPI, COD, CC).
Gender	Categorical	Input	Customer gender identity (Male, Female).
HourSpendOnApp	Numerical (Float)	Input	Average hours spent browsing the application daily.
NumberOfDeviceRegistered	Numerical (Int)	Input	Total registered devices connected to customer account (1 to 6).
PreferedOrderCat	Categorical	Input	Dominant purchasing category (Mobile, Fashion, Grocery, etc.).
SatisfactionScore	Numerical (Int)	Input	Customer satisfaction rating on a 1 (Lowest) to 5 (Highest) scale.
MaritalStatus	Categorical	Input	Marital status of customer (Single, Married, Divorced).
NumberOfAddress	Numerical (Int)	Input	Total registered shipping addresses on account.
Complain	Binary Integer	Input	Customer raised a complaint within the past month (1 = Yes, 0 = No).
OrderAmountHikeFromlastYear	Numerical (Float)	Input	Percentage growth in spending compared to prior year.
CouponUsed	Numerical (Float)	Input	Total promotional discount coupons redeemed in the past year.
OrderCount	Numerical (Float)	Input	Total count of completed purchase transactions.
DaySinceLastOrder	Numerical (Float)	Input	Recency in days since customer's most recent order.
CashbackAmount	Numerical (Float)	Input	Average cashback reward (\$) credited per order.

4. FROM-SCRATCH DESCRIPTIVE STATISTICAL ENGINE (TASK B2)

To profile dispersion and central tendencies without relying on automated library functions, statistical parameters were calculated manually across all 13 numerical inputs:

$$\mu = (1/n) \sum X_i \quad | \quad \sigma^2 = (1/n) \sum (X_i - \mu)^2 \quad | \quad \sigma = \sqrt{\sigma^2} \quad | \quad Range = X_{max} - X_{min}$$

Feature	Mean	Median	Mode	Min	Max	Range	Variance (σ^2)	Std Dev (σ)
Tenure	10.25	9.00	1.00	0.00	61.00	61.00	73.59	8.58
WarehouseToHome	15.68	14.00	9.00	5.00	127.00	122.00	73.48	8.57
HourSpendOnApp	2.92	3.00	3.00	0.00	5.00	5.00	0.52	0.72
NumberOfDeviceRegistered	3.68	4.00	4.00	1.00	6.00	5.00	1.03	1.02
NumberOfAddress	4.20	3.00	2.00	1.00	21.00	20.00	6.63	2.58
OrderCount	3.01	2.00	2.00	1.00	16.00	15.00	8.74	2.96
CouponUsed	1.74	1.00	1.00	0.00	15.00	15.00	3.49	1.87
DaySinceLastOrder	4.54	3.00	3.00	0.00	46.00	46.00	13.50	3.67

Feature	Mean	Median	Mode	Min	Max	Range	Variance (σ^2)	Std Dev (σ)
CityTier	1.66	1.00	1.00	1.00	3.00	2.00	0.84	0.92
SatisfactionScore	3.07	3.00	3.00	1.00	5.00	4.00	1.92	1.39
OrderAmountHikeFromlastYear	15.70	15.00	13.00	11.00	26.00	15.00	13.52	3.68
Complain	0.28	0.00	0.00	0.00	1.00	1.00	0.20	0.45
CashbackAmount	177.04	163.23	149.36	0.00	324.99	324.99	2424.58	49.24

5. DATA QUALITY AUDITING, MISSINGNESS & CLEANING

5.1 Missing Value Diagnostics (Task C1) & Justified Imputation (Task C2)

Missing value scanning identified **1,478 total null entries** across exactly 7 numerical columns (~4.40% to 5.26% missingness per feature). Custom imputation rules were applied based on distributional skewness and feature semantics:

- **Median Imputation (Skewed Continuous Data):** Applied to Tenure, WarehouseToHome, OrderAmountHikeFromlastYear, and DaySinceLastOrder because extreme values (e.g., WarehouseToHome max = 127 km) would distort the sample mean.
- **Mean Imputation (Symmetric Distribution):** Applied to HourSpendOnApp, which displays an approximately normal bell distribution with mean (2.92) $\approx$ median (3.00).
- **Mode Imputation (Discrete Counts):** Applied to discrete count variables CouponUsed and OrderCount to ensure imputed values reflect valid, discrete transaction integers.

Feature Name	Missing Count	Missing %	Distributional Shape	Selected Imputation Metric	Pre-Impute Nulls	Post-Impute Nulls
DaySinceLastOrder	237	5.26%	Right-Skewed Tail (Max = 46.0)	Median (3.0)	237	0
Tenure	218	4.84%	Heavy Right Skew (Max = 61.0)	Median (9.0)	218	0
OrderAmountHikeFromlastYear	212	4.71%	Moderately Right-Skewed	Median (15.0)	212	0
WarehouseToHome	206	4.57%	Heavy Outlier Tail (Max = 127.0)	Median (14.0)	206	0
CouponUsed	205	4.55%	Discrete Count Feature	Mode (1.0)	205	0
OrderCount	202	4.48%	Discrete Count Feature	Mode (2.0)	202	0
HourSpendOnApp	198	4.40%	Symmetric (Mean = 2.92, Med = 3.0)	Mean (2.92)	198	0

5.2 Duplicate Verification (Task D1) & Categorical Cleanup (Task D2)

Duplicate Detection Engine (Task D1): A custom hash-set tracker evaluated unique CustomerID occurrences:

Validation Metric	Value
Initial Training Records	4,504
Duplicate Primary Keys	0
Retained Observations	4,504

Inconsistent Data Resolution (Task D2): Standardized naming inconsistencies, whitespace errors, and multi-alias representations:

- PreferredLoginDevice: Collapsed Mobile Phone → Phone (3 unique → 2: *Phone, Computer*).
- PreferredPaymentMode: Standardized CC → Credit Card, COD → Cash On Delivery, E wallet → E Wallet (7 → 5).

Outcome: 100% record integrity verified; zero duplicate data corruption.

- PreferredOrderCat: Collapsed Mobile Phone → Mobile (6 → 5).
- Validated all numerical boundaries (0 negative/impossible values).

6. CATEGORICAL ENCODING, OUTLIER ANALYSIS & SCALING

6.1 From-Scratch Encoding Implementations (Tasks E1 & E2)

- **Task E1 (Label Encoding):** Binary variable Gender mapped via custom dictionary: {'Female': 0, 'Male': 1} (Female: 1,802, Male: 2,702). Label encoding was restricted to binary variables to prevent artificial numeric ordering in nominal multiclass variables.
- **Task E2 (One-Hot Encoding):** Implemented from scratch for 4 multiclass nominal columns (PreferredLoginDevice, PreferredPaymentMode, PreferredOrderCat, MaritalStatus), generating 15 binary dummy indicator features and expanding feature dimensionality from 20 to 36.

6.2 Outlier Analysis: IQR vs. Z-Score Comparison (Tasks F1 & F2)

Outliers were detected from scratch using Interquartile Range ($Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR$) and Z-Score ($|Z| > 3.0$).

Feature Name	Q1	Q3	IQR	IQR Bounds [Lower, Upper]	IQR Outliers Detected	Z-Score Outliers ($ Z > 3.0$)
OrderCount	1.00	3.00	2.00	[-2.00, 6.00]	567	148
CouponUsed	1.00	2.00	1.00	[-0.50, 3.50]	504	88
CashbackAmount	145.80	196.19	50.39	[70.22, 271.77]	351	10
NumberOfDeviceRegistered	3.00	4.00	1.00	[1.50, 5.50]	310	0
DaySinceLastOrder	2.00	7.00	5.00	[-5.50, 14.50]	52	37
OrderAmountHikeFromlastYear	13.00	18.00	5.00	[5.50, 25.50]	27	0
HourSpendOnApp	2.00	3.00	1.00	[0.50, 4.50]	6	3
Tenure	3.00	15.00	12.00	[-15.00, 33.00]	3	3
WarehouseToHome	9.00	20.00	11.00	[-7.50, 36.50]	2	2
NumberOfAddress	2.00	6.00	4.00	[-4.00, 12.00]	2	2
CityTier / SatisfactionScore / Complain	Bounded Discrete Inputs				0	0
Total Outlier Instances	—				1,824	293

Methodological Insight: Why Outlier Counts Differ

IQR (1,824 outliers) uses ranking percentiles, robustly isolating extreme tails in narrow-spread discrete columns. Z-Score (293 outliers) relies on sample variance; extreme outliers inflate standard deviation σ , widening the $\pm 3\sigma$ threshold and creating a masking effect on moderate anomalies.

6.3 Outlier Treatment via Non-Destructive Transformation (Tasks F3 & G)

Justification: Dropping 1,824 rows would purge ~40.5% of the dataset, eliminating valid high-value customers. Instead of deletion or artificial winsorization capping, a natural log transformation was applied from scratch: $X' = \ln(X + 1)$ on CashbackAmount and OrderAmountHikeFromlastYear. This compressed positive skewness while preserving exact relative ordering and variance structure.

6.4 Feature Scaling: Normalization vs. Standardization (Tasks H1 & H2)

Implemented from-scratch scaling functions: Min-Max Normalization ($X' = (X - X_{min}) / (X_{max} - X_{min})$) bounded features strictly into $[0, 1]$, and Z-Score Standardization ($Z = (X - \mu) / \sigma$) centered features at $\mu = 0.0, \sigma = 1.0$.

7. DATA LEAKAGE PREVENTION & STATISTICAL FEATURE SELECTION

7.1 Leakage Architecture (Task K & M9)

Fitting preprocessing or performing supervised feature selection on the combined dataset allows test-set information to leak into the training model, creating inflated validation scores that fail in production.

Pipeline Leakage Isolation Rules:

- Train-Test Split executed *first* (80/20 index split).
- Imputation statistics (μ_{train}, Med_{train}) calculated *strictly from training data*.
- Scaling bounds (X_{min}, X_{max}, σ) learned from train set and applied to test set.
- Supervised statistical tests (Chi-Square, ANOVA) evaluated *solely* against training target labels.

7.2 Method Selection Matrix (Task M6)

Method	Input Type	Target	Mathematical Purpose
Variance Threshold	Numerical / OHE	None	Detect constant / quasi-constant features.
Pearson Corr (r)	Numerical	Target	Measure linear relationship & redundancy.
Chi-Square (χ^2)	Categorical	Categorical	Test categorical independence against churn.
ANOVA F-Test	Numerical	Categorical	Compare between-group vs. within-group variance.

7.3 Multi-Criteria Feature Selection Decision Table (Tasks M1–M7)

$$r = \Sigma((x - \bar{x})(y - \bar{y})) / \sqrt{[\Sigma(x - \bar{x})^2 \Sigma(y - \bar{y})^2]} \quad | \quad \chi^2 = \Sigma (O - E)^2 / E \quad | \quad F = [SSB / (k - 1)] / [SSW / (N - k)]$$

Feature Name	Data Type	Variance	Pearson (r)	ANOVA F	Chi-Sq (χ^2)	Decision	Empirical Justification
Tenure	Numerical	73.59	-0.342	591.20	—	KEEP	Strongest predictor overall; high tenure substantially lowers churn likelihood.
Complain	Binary Num	0.20	+0.250	291.98	274.19	KEEP	Complaining customers churn at 31.5% vs. 11.0% non-complainants ($\chi^2 = 274.19$).
DaySinceLastOrder	Numerical	13.50	-0.160	116.83	—	KEEP	Highly significant recency indicator; prolonged purchase inactivity drives churn.
CashbackAmount	Numerical	2424.58	-0.154	108.45	—	KEEP	Higher cashback provides financial incentive for platform retention.
SatisfactionScore	Numerical	1.92	+0.105	49.97	—	KEEP	Significant variance across churned and retained customer satisfaction ratings.
NumberOfDeviceRegistered	Numerical	1.03	+0.108	52.79	—	KEEP	Significant F-score; high registered devices correlate with shared accounts & churn.

Feature Name	Data Type	Variance	Pearson (r)	ANOVA F	Chi-Sq (χ^2)	Decision	Empirical Justification
CityTier	Numerical	0.84	+0.085	32.70	—	KEEP	Tier 3 urban centers exhibit distinct delivery friction and churn rates.
WarehouseToHome	Numerical	73.48	+0.070	22.15	—	KEEP	Delivery distance impacts fulfillment latency; ANOVA $F > 3.84$ critical threshold.
NumberOfAddress	Numerical	6.63	+0.044	8.78	—	KEEP	Significant variation across churn groups ($F = 8.78 > F_{\text{critical}}$).
OrderCount	Numerical	8.74	-0.029	3.89	—	KEEP	Exceeds critical F-score threshold (3.84); retained as baseline transaction volume.
MaritalStatus	Categorical	—	—	—	98.42	KEEP	High statistical dependency; single individuals exhibit elevated churn rates.
PreferredOrderCat	Categorical	—	—	—	84.15	KEEP	Product category significantly dictates repeat purchasing habits.
PreferredPaymentMode	Categorical	—	—	—	57.30	KEEP	Payment method friction (e.g. COD vs. Credit Card) influences retention.
PreferredLoginDevice	Categorical	—	—	—	24.60	KEEP	Statistically dependent on churn target ($\chi^2 = 24.60 >$ critical value, $p < 0.001$).
Gender	Categorical	0.24	+0.029	3.85	4.12	KEEP	Statistically significant under Chi-Square ($\chi^2 = 4.12 > 3.841$, $df = 1$, $p < 0.05$).
HourSpendOnApp	Numerical	0.52	+0.019	1.66	—	REMOVE	ANOVA $F (1.66) < 3.84$ critical threshold; Pearson $r \approx 0$. No discriminatory signal.
CouponUsed	Numerical	3.49	-0.008	0.32	—	REMOVE	ANOVA $F (0.32)$ fails significance; near-zero correlation with customer churn.
OrderAmountHikeFromlastYear	Numerical	13.52	-0.0002	0.0001	—	REMOVE	$F\text{-score} \approx 0$ and $r \approx 0$; statistically independent of target churn variable.

8. FINAL PIPELINE RESULTS & COMPARATIVE AUDIT (TASK N)

8.1 Feature Dimensionality Flow (Task M8)

Pipeline Execution Stage	Feature Count	Stage Definition
1. Original Dataset	19 Features	18 input features + 1 target (Churn)

8.2 Complete Preprocessing Comparison Table (Task N)

Parameter	Before Preprocessing	After Preprocessing
Records	5,630	4,504

Pipeline Execution Stage	Feature Count	Stage Definition	Parameter	Before Preprocessing	After Preprocessing
2. After Preprocessing	35 Features	13 numerical + 1 label encoded + 21 OHE dummy features	Features	19	35
3. After Feature Selection	26 Features	Statistically validated features (3 uninformative inputs removed)	Missing Values	1,478	0
			Duplicate Records	0	0
			Categorical Features	7	0
			Outliers (IQR Method)	1,824	1,824
			Outliers (Z-Score Method)	293	293
			Selected Features	18	26

Detailed Interpretation of Task N Audit Parameters:

- **Records (5,630 → 4,504):** Preprocessing and modeling transformations are executed strictly on the 80% training split (4,504 records) to prevent test-set data leakage.
- **Features (19 → 35):** Raw columns expanded as 7 multi-class and binary categorical variables were transformed into 1 label encoded column and 21 One-Hot Encoded binary indicators.
- **Missing Values (1,478 → 0):** Fully resolved across all 7 numerical features using skewness-aligned Median, Mean, and Mode imputation without deleting rows.
- **Categorical Features (7 → 0):** 100% of raw text categorical features were converted into machine-readable numerical representations.
- **Outliers (1,824 IQR / 293 Z-Score):** 100% retained and treated via non-destructive $\ln(X+1)$ transformation, avoiding sample truncation while reducing positive skewness.
- **Selected Features (18 → 26):** After expanding to 35 features, statistical elimination of 3 non-significant inputs (``HourSpendOnApp``, ``CouponUsed``, ``OrderAmountHikeFromlastYear``) finalized the pipeline at 26 validated model-ready features.

9. KEY ANALYTICAL FINDINGS & PRACTICAL INSIGHTS

1. **Customer Complaints are the Dominant Churn Driver:** Customers filing complaints exhibit a **31.5% churn rate** versus **11.0%** for non-complainants ($\chi^2 = 274.19$, ANOVA $F = 291.98$). Automated escalation workflows for customer complaints represent the single highest-ROI retention initiative.
2. **Tenure Protects Against Churn:** Customer tenure displayed the strongest overall statistical relationship (ANOVA $F = 591.20$, Pearson $r = -0.342$). Newly onboarded users are most vulnerable; dedicated onboarding campaigns in months 1–3 are crucial.
3. **Linear Correlation Alone is Insufficient:** Low linear correlation features such as WarehouseToHome ($r = +0.070$) and CashbackAmount ($r = -0.154$) demonstrated strong non-linear significance under ANOVA ($F = 22.15$ and 108.45). Feature selection must incorporate multi-test statistical validation.
4. **Justified Elimination of Irrelevant Features:** HourSpendOnApp ($F = 1.66$), CouponUsed ($F = 0.32$), and OrderAmountHikeFromlastYear ($F = 0.0001$) were proven statistically independent of churn and safely eliminated, reducing model complexity and training latency.

Conclusion

By combining from-scratch mathematical preprocessing with leak-free statistical feature selection, the raw e-commerce dataset was transformed into a robust, clean, and statistically validated 26-feature dataset primed for high-performance machine learning classification.